

20251119EB_MN (NCT, IF)

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Status: In progress
Entry Created On: 10 Dec 2025 23:54:17 UTC
Entry Last Modified: 10 Mar 2026 16:04:12 UTC
Export Generated On: 01 Sep 2026 18:31:08 UTC

- 2025年11月19日 星期三
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年11月21日 星期五
- T2 -> T3
- 2025年11月23日 星期日
- T3 -> T4
- 2025年11月25日 星期二
- +4mL T4
- 2025年11月26日 星期三
- T4 -> T5
- 2025年11月28日 星期五
- T5 -> T6 (~14mL for over-weekend)
- 2025年12月1日 星期一
- T6 -> T6
- 2025年12月3日 星期三
- T6 -> T7
Dissociation
 1. Collect EB and wash plate with PBS
 2. Let EB settle with gravity in 15mL tube
 3. Add 800uL 0.025%trypsin w/ 50U/mL DNaseI
 4. Pipette *5 and add additional 800uL trypsin --> incubate at 37C
 5. pipette at 5, 10, 15, 20 and 25min
 6. At 5min, transfer 800uL supernatent to HI-FBS and add 400uL more trypsin to 15mL tube
 7. At 10 min add 500uL trypsin
 8. At 15min, transfer 500uL to FBS
 9. At 20 min, transfer 800uL to FBS +500uL trypsin (still many EBs visible)

10. At 25min, pipette and tansfer all to HI-FBS (6mL in total. Mix the previous premium HI-FBS and the current HI-FBS used in the lab)
11. Filter through 40um cell strainer
12. centrifuge at 200xg, 3.5min

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P6	3	

Table1			^
	A	B	
1		volume (mL)	
2	trypsin	2.5	
3	HI-FBS/CTWM	6	

Table3									/
	A	B	C	D	E	F	G	H	
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	
2	1.76	100	2	3.52	A	24	220	16	

Well1							^			
	1	2	3	4	5	6				
A			110	111	no virus					
B			12uL in	18uL in						
C			total	total						
D			no virus							

2025年12月4日 星期四

Change T7 -> T7

2025年12月6日 星期六

T7 -> T8 (I aspirate the media and cells with 111 detached a bit)

2025年12月8日 星期一

1/2 T8 -> 1/2 T8

2025年12月10日 星期三

1/2 T8 -> 1/2 T8

2025年12月12日 星期五

1/2 T8 -> 1/2 T8

2025年12月14日 星期日

1/2 T8 -> 1/2 T8

2025年12月15日 星期一

condition#			^
	A	B	
1	condition#		
2	1	2.5 mM > 2.5 mM 1h > 2.5 mM 3h > 2.5 mM 1h > 2.5 mM 3h > 2.5 mM 1h -> 2.5mM	
3	2	2.5 mM > 0.15 (1h) > 2.5 mM 3h > 0.15 (1h) > 2.5 mM 3h > 0.15 (1h) -> 0.15mM	
4	3	2.5 mM > 0.15 mM 9h -> 0.15mM	
5	4	2.5mM 1h	
6	5	0.15mM 1h	
7	6	2.5mM 1h without pyruvate	
8	7	0.15mM 1h without pyruvate	

Conditions for NCT imaging							^
	1	2	3	4	5	6	
A			1	1			
B			2	2			
C			3	3			
D							

time table for NCT imaging																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1			0h		1h			4h		5h		8h		9h			24h
2	condition#	condition	media	time	imaging	media after imaging	time	media	time	media	time	media	time	imaging	media after imaging	time	imaging on 12/16
3		110															
4	1	2.5mM glucose repetitive change	2.5mM gluc	11:00	12:00	2.5mM gluc	12:20	2.5mM gluc	15:00	2.5mM gluc	16:30 I forgot to change this at 16:00	2.5mM gluc	19:00	20:00	2.5mM gluc	20:18	12:00
5	2	0.15mM glucose repetitive change	0.15mM gluc	11:30	12:56 (the images were out of focus twice --> delayed...) The current image was activated for three rounds	2.5mM gluc	13:15	0.15mM gluc	15:30	2.5mM gluc	16:30	0.15mM gluc	19:30	20:37	0.15mM gluc	20:50	12:30
6	3	0.15mM glucose constant	0.15mM gluc	12:00	13:21	NA		NA		NA		NA		21:00	0.15mM gluc	21:20	13:05
7		111															
8	1	2.5mM glucose repetitive change	2.5mM gluc	12:30	13:55	2.5mM gluc	14:10	2.5mM gluc	16:30	2.5mM gluc	17:30	2.5mM gluc	20:28	21:30	2.5mM gluc	21:50	13:40
9	2	0.15mM glucose repetitive change	0.15mM gluc	13:15	14:17	2.5mM gluc	14:35	0.15mM gluc	17:00	2.5mM gluc	18:00	0.15mM gluc	21:00	22:00	0.15mM gluc	22:20	14:03
10	3	0.15mM glucose constant	0.15mM gluc	13:42	14:42	NA		NA		NA		NA		22:23	0.15mM gluc	22:50	14:32

fileID_treatment_NCT						
	A	B	C	D	E	F
1	image	reporter	treatment	time	type	
2	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	2.5mM gluc	1h	continuous	
3	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	2.5mM gluc	1h	continuous	
4	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.15mM gluc	1h	recurrent	
5	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0.15mM gluc	1h	recurrent	
6	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0.15mM gluc	1h	no-change	
7	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	0.15mM gluc	1h	no-change	
8	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	2.5mM gluc	9h	continuous	
9	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	2.5mM gluc	9h	continuous	
10	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	0.15mM gluc	9h	recurrent	
11	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	0.15mM gluc	9h	recurrent	
12	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	0.15mM gluc	9h	no-change	
13	com_110_MN_no-glu_50%_1s_30s_60x_0012	110	0.15mM gluc	9h	no-change	position 1 is out of focus
14	com_110_MN_no-glu_50%_1s_30s_60x_0013	110	2.5mM gluc	24h	continuous	
15	com_110_MN_no-glu_50%_1s_30s_60x_0014	110	2.5mM gluc	24h	continuous	
16	com_110_MN_no-glu_50%_1s_30s_60x_0015	110	0.15mM gluc	24h	recurrent	
17	com_110_MN_no-glu_50%_1s_30s_60x_0016	110	0.15mM gluc	24h	recurrent	
18	com_110_MN_no-glu_50%_1s_30s_60x_0017	110	0.15mM gluc	24h	no-change	
19	com_110_MN_no-glu_50%_1s_30s_60x_0018	110	0.15mM gluc	24h	no-change	
20	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	2.5mM gluc	1h	continuous	
21	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	2.5mM gluc	1h	continuous	
22	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.15mM gluc	1h	recurrent	
23	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0.15mM gluc	1h	recurrent	
24	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0.15mM gluc	1h	no-change	
25	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	0.15mM gluc	1h	no-change	
26	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	2.5mM gluc	9h	continuous	
27	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	2.5mM gluc	9h	continuous	
28	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	0.15mM gluc	9h	recurrent	
29	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	0.15mM gluc	9h	recurrent	
30	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	0.15mM gluc	9h	no-change	
31	com_111_MN_no-glu_50%_1s_30s_60x_0012	111	0.15mM gluc	9h	no-change	
32	com_111_MN_no-glu_50%_1s_30s_60x_0013	111	2.5mM gluc	24h	continuous	
33	com_111_MN_no-glu_50%_1s_30s_60x_0014	111	2.5mM gluc	24h	continuous	
34	com_111_MN_no-glu_50%_1s_30s_60x_0015	111	0.15mM gluc	24h	recurrent	
35	com_111_MN_no-glu_50%_1s_30s_60x_0016	111	0.15mM gluc	24h	recurrent	
36	com_111_MN_no-glu_50%_1s_30s_60x_0017	111	0.15mM gluc	24h	no-change	
37	com_111_MN_no-glu_50%_1s_30s_60x_0018	111	0.15mM gluc	24h	no-change	

*20260310 updated the type annotation: repetitive change to the same concentration: **constant**; no change at all: **no-change**

2025年12月16日 星期二

Prepare media without pyruvate
2.5mM gluc: 4.55uL 1.1M gluc in 2000uL media
0.15mM gluc: 120uL 2.5uL gluc media in 1880uL no gluc media

Prepare 4%PFA in PBS: $250 \times 10 = 2500 \rightarrow 3000$
375uL 32%PFA in 2625uL PBS

1. Treat cells from 15:10-15:20
2. Wash cells with cold PBS at 16:10
3. Fix cells with 4% PFA for 10 min at 16:10-16:20
4. Wash with PBS
5. Permeabilization with 0.2% triton for 15min 16:25-16:40
6. PBS wash
7. Blocking with **10% FBS** for 1 hour 16:50-20:00
8. Primary antibodies overnight

Conditions for IF treatment							^
	1	2	3	4	5	6	
A					4	4	
B					5	5	
C					6	6	
D			4	5	7	7	

Antibody layout for IF staining							^
	1	2	3	4	5	6	
A					A	B	
B					A	B	
C					B	A	
D			B	B	B	A	

Note1: B6 and D6 didn't look well after fixation (I forgot to check them prior to the treatment). Neurites broke a lot.

Note2: I accidentally add around 2-4uL AntibodyA into D5. So I remove the solution and re-applied AntibodyB. However, a part of cells detached after that.

IF list:

RANGAP1 (C-31, Rb)
Ran (146, Mouse)
TDP43 (33, Rb)
Importin-b1 (147, M)

Table7												
	A	B	C	D	E	F	G	H	I	J	K	L
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	750	1.5	488	B	1000	1500	1.5
3	A	147	Importin-b1	m	1000	750	0.75	594	O	1000	1500	1.5
4	B	C-31	RANGAP1	Rb	500	1080	2.16	488	B			
5	B	146	Ran	m	500	1080	2.16	594	O			
6	C	C-31	RANGAP1	Rb	500	360	0.72					
7	C	147	Importin-b1	m	1000	360	0.36					

2025年12月17日 星期三

- 1. wash with 5 min PBS *3
- 2. 2nd antibody in 10% FBS for 1h RT. 14:00-15:00
- 3. wash with 5 min PBS *3
- 4. check staining
- 5. DAPI staining for 10min
- 6. PBS wash *3
- 7. Store in the cold room

2025年12月19日 星期五

Imaging IF

Note: cells in many wells detached

- **A5:** some cells detached a bit. I noticed the first image include some detached cluster. --> retook the image again (_001). Pick more cluster around the edge to make the them more comparable to other groups.
- **B5:** a bit part of cells detached. Some cells still attached at the edge of the well.
- **C6:** most cells attached well.
- **D6:** neurites are beading. Cells look very unhealthy. Only few cells survived. --> only took 6 fields for example since cells are not healthy.
- **D3:** cells look okay. Some detached a bit
- **D4:** Most cells detached... ---> can't find good cells. Didn't take images.
- **B6:** A lot of cells detached as well.... --> try to find 8 fields at the edge
- **C5:** cells look okay
- **D5:** most cells detached. Try to find 12 fields.
- **A6:** most cells died/neurites beaded. didn't take images.

SetA OC 60x,704x704, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	DIC			BF
3	405	10	50	DAPI
4	488	10	100	TDP43
5	561	30	500	Importin-b1
6	640	30	300	Halodye (TDP43-Halo)
7	z-stack	1um/step	15um	17 steps in total

SetB OC 60x,704x704, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	DIC			BF
3	405	10	50	DAPI
4	488	10	200	RANGAP1
5	561	25	300	Ran
6	640	30	300	Halodye (TDP43-Halo)
7	z-stack	1um/step	15um	17 steps in total

fileID_treatment_IF					
	A	B	C	D	E
1	image	treatment	treat_time	media	note
2	A5_488-TDP43_561-Impb1_640-Halo_405-DAPI	2.5mM gluc	1h	full	include some detached clusters
3	A5_488-TDP43_561-Impb1_640-Halo_405-DAPI_001	2.5mM gluc	1h	full	
4	B5_488-TDP43_561-Impb1_640-Halo_405-DAPI	0.15mM gluc	1h	full	
5	C6_488-TDP43_561-Impb1_640-Halo_405-DAPI	2.5mM gluc	1h	no-pyruvate	
6	D6_488-TDP43_561-Impb1_640-Halo_405-DAPI	0.15mM gluc	1h	no-pyruvate	
7	D3_488-RANGAP1_561-Ran_640-Halo_405-DAPI	2.5mM gluc	1h	full	
8	B6_488-RANGAP1_561-Ran_640-Halo_405-DAPI	0.15mM gluc	1h	full	
9	C5_488-RANGAP1_561-Ran_640-Halo_405-DAPI	2.5mM gluc	1h	no-pyruvate	
10	D5_488-RANGAP1_561-Ran_640-Halo_405-DAPI	0.15mM gluc	1h	no-pyruvate	

analysis: exclude A5_488-TDP43_561-Impb1_640-Halo_405-DAPI